

Recording a route without satellites: vibration-derived speed and its failure modes

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Abstract

A phone can record where it has been without satellite positioning, by reading speed from how the vehicle shakes and direction from its motion sensors. We describe the method as shipped in a consumer iOS workout application, and report its accuracy on twenty-three instrumented journeys by car, motorcycle and on foot. Each was recorded with GPS running alongside, used only as a reference: we call that the *real* speed, and what the sensors produced the *estimated* speed. Driving fast and steadily with the phone resting in the vehicle, estimated distance is within 3.8% over 15 km; in slow stop-start traffic the same system reads 20–60% long. Direction needs no satellite reference either: the angle the phone sits at is learned from the sideways acceleration of turns, and replayed over 61 recordings the saved route points within 30° of the real direction for 72% of riding, against 43% for the phone’s compass alone; walking with the phone in a pocket, direction from the steps is within 30° for 93% of the time. Part of that gap is a property of the signal — the estimate reads one to two km/h high at any speed, which is 2% of a motorway pace and 40% of a crawl — but most of it turned out not to be. The store evicts from its most crowded speed bin so that rare fast examples survive, and that policy alone leaves it holding 5% of its examples below 5 km/h where the riding produces 24%. Replaying the estimator against a store rebuilt under that rule reproduces the shipped application’s error on six instrumented rides; the same estimator on an unrebuilt store does not. Undoing the imbalance at lookup takes mean distance error from 20% to 8% (§4.3). We give equal space to eighteen approaches that were implemented, measured and rejected. Two failed for one reason. An accelerometer cannot tell gravity apart from the vehicle pushing it — long known in attitude estimation — and on this platform the sensor fusion that resolves that ambiguity is the vendor’s, cannot be replaced, and quietly removes the very signal that aircraft dead reckoning and takeoff detection rely on. We report how much that costs, and correct an earlier claim that it also rules out direction from acceleration.

1 Introduction

Satellite positioning fails where routes are most often wanted: tunnels, underground car parks, dense urban canyons, and aircraft cabins. The standard response is inertial dead reckoning — integrate acceleration twice — which is well known to diverge, since accelerometer bias is indistinguishable from sustained real acceleration and its double

integral grows without bound.

An alternative exists for wheeled vehicles. Road and engine vibration scale with speed, so a spectral signature of vertical acceleration can be mapped to speed directly, without integration and therefore without accumulation. CarSpeedNet [1] reports RMSE 1.8 m/s and MAE 0.72 m/s from 13.2 hours of accelerometer-only driving, at the same 4-second window length we arrive at independently. Its 0.5-hour test holdout is drawn from that same collection, so the reported figure describes journeys much like the ones it learned from.

This paper describes such a system built into a shipping application, and — more usefully — reports what happens outside the distribution, on a phone that is sometimes mounted, sometimes pocketed, sometimes in a hand, in a car, on a motorcycle, and once in an aircraft.

Contributions.

1. An out-of-distribution evaluation of vibration-derived speed across carry positions and vehicle types, on one device and one user, with per-regime error.
2. The measured cost of the specific-force ambiguity on a consumer platform: two distinct approaches defeated, with magnitudes.
3. A cache-policy bias: the eviction rule that protects rare fast examples silently reshapes the training distribution, and accounts for most of the low-speed over-read (§4.3).
4. Direction without GPS — a heading that cannot go singular, the carry angle from cornering, and walking direction from the step axis — replayed over every recording with the routes drawn against GPS for all 61 (§4.4, Appendix A).
5. Eighteen negative results reported in full, including two that look correct in aggregate and fail per-query.
6. Three sensor-platform defects that silently invalidated weeks of data, offered as evidence that platform semantics dominate modelling error in this setting.

2 Related work

Speed from vibration. The closest work is CarSpeedNet [1], a CNN taking three-axis smartphone accelerometer input, reporting RMSE 1.8 m/s and MAE 0.72 m/s over 13.2 hours of driving. Related approaches estimate speed from chassis vibration with CNNs [2], infer position directly from vehicle vibration with an IMU [3], and use learned momentary speed to constrain strapdown inertial navigation [4]. Outside the inertial domain, vehicle speed has been estimated from tyre-road noise, where band energies at 1–4 kHz give 1.4% average error [5] and the underlying physics — tread-block passage at $f = NV/L$ — gives a direct speed dependence.

Our estimator is deliberately simpler than these: a k -NN regressor over log-spectral band energies, chosen because it can *decline* to answer when nothing resembling the query has been observed, which a trained network cannot. The contribution here is not the estimator but the evaluation regime.

Published accuracy is usually measured on familiar data. CarSpeedNet’s 0.5-hour holdout comes from the same 13.2-hour collection as its training data. This matters: we measure +1.0 to +1.7 km/h bias on a store built from matched journeys and +6.8 km/h on the same device once the store contains other carry positions and vehicles — a degradation invisible to a random split. The transportation-mode literature has confronted this more directly; the Sussex–Huawei Locomotion dataset [6] was assembled specifically to enable reproducible cross-user evaluation, with benchmark F1 near 0.93 for eight-class mode recognition [7]. No equivalent public dataset exists for *speed* regression across carry positions.

Inertial dead reckoning. That double integration of consumer MEMS output is unusable is well established: laboratory characterisation of smartphone IMUs reports position error exceeding 1 km within 100 s from pure inertial propagation [8], with error growth dominated by uncompensated bias and varying substantially between handsets. Learned approaches to inertial odometry [9] and the broader survey of deep inertial positioning [10] attack this by replacing integration with regression, as we do. Standard mitigations — zero-velocity updates [11] and map matching [12] — bound rather than remove the drift.

Heading and carry position. Pedestrian dead reckoning divides into step detection, heading estimation and position propagation [13], and heading is consistently the limiting term. Carry position is known to dominate: measured heading standard deviations of 6.22° , 6.82° and 14.68° for holding, calling and swinging respectively, with corresponding final position errors of 2.11, 2.34 and 4.5 m [14]. Our β term is the vehicular analogue of the carry-mode misalignment that literature addresses for pedestrians. An earlier draft of this paper reported that β cannot be recovered without an external reference. That was wrong: a vehicle’s turns reveal it (§4.4).

The specific-force ambiguity. An accelerometer measures specific force and cannot separate gravity from sustained linear acceleration. This is standard, and attitude-estimation work explicitly compensates for it, either by estimating disturbance acceleration adaptively [15] or by weighting the accelerometer correction by how far its magnitude departs from g [16]. What we add is the downstream consequence on a platform where the fusion is not ours to modify: the vendor-supplied attitude and gravity-removed acceleration are the only interface, and their handling of sustained acceleration silently defeats two otherwise reasonable designs (§5.2).

3 Method

3.1 Signature

Vertical acceleration $x[n]$ is sampled at 50 Hz into a ring buffer of $N = 256$ samples (4.0 s), windowed and transformed:

$$w[n] = \frac{1}{2} - \frac{1}{2} \cos\left(\frac{2\pi n}{N-1}\right) \quad (1)$$

$$X[k] = \sum_{n=0}^{N-1} (x[n] - \bar{x}) w[n] e^{-2\pi i k n / N} \quad (2)$$

Energy is accumulated into nine logarithmically spaced bands bounded at 0.4, 0.8, 1.6, 2.5, 4, 6, 9, 13, 18 and 24 Hz:

$$f_b = \ln\left(\sum_{k \in b} |X[k]|^2 + \epsilon\right), \quad b = 1 \dots 9 \quad (3)$$

Two time-domain terms complete an eleven-dimensional signature:

$$f_{10} = \ln \sigma(x), \quad f_{11} = \ln\left(\frac{1}{N-1} \sum_n |x[n] - x[n-1]|\right) \quad (4)$$

The window length was measured rather than assumed. At 4 s the held-out bias is +1.0 km/h at 2–10 km/h; doubling to 8 s raises it to +6.0, because the window then straddles changes in speed. Finer low-frequency banding changed nothing.

3.2 Speed

No functional form is imposed. The system stores $(\mathbf{f}, v)$ pairs whenever GPS supplies a speed, and answers by locality. With running feature means μ_i and variances σ_i^2 ,

$$d^2(\mathbf{f}, \mathbf{g}) = \sum_{i=1}^{11} \left(\frac{f_i - \mu_i}{\sigma_i} - \frac{g_i - \mu_i}{\sigma_i} \right)^2 \quad (5)$$

$$\hat{v} = \frac{\sum_{j=1}^K w_j v_j}{\sum_{j=1}^K w_j}, \quad w_j = \frac{1}{d_j^2 + \epsilon} \quad (6)$$

with $K = 12$, where π_j corrects for how over- or under-represented a neighbour's speed is in the store relative to how often that speed was actually travelled:

$$\pi(v) = \text{clamp}\left(\frac{P_{\text{ridden}}(\text{bin}(v))}{P_{\text{store}}(\text{bin}(v))}, 0.05, 20\right), \quad w_j = \frac{\pi(v_j)}{d_j^2 + \epsilon} \quad (7)$$

P_{ridden} is a decaying histogram of every observation as it was learned, over 2 km/h bins, with a half-life of about 1400 observations; P_{store} is the same histogram of what the store currently holds. The two differ because of the eviction policy, not because of anything

about the signal, and §4.3 shows that difference accounts for most of the low-speed over-read. Both are 1 on a fresh install, where the estimator reduces to the unweighted form above. Crucially the estimator is allowed to decline:

$$\min_j d_j^2 > 4 \implies \text{no answer} \quad (8)$$

Error scales directly with match distance — MAE 5.3 km/h below $d^2 = 1.03$, rising to 16.9 in the range 2.12–4.36 — so refusing a distant match removes the worst readings.

3.3 Regime separation

A store holding several vehicles holds several mutually incompatible mappings. Measured directly: a ground-trained model reported 19 km/h at a 900 km/h cruise, because cruise is *quieter* than taxiing and the slowest stored signature was the closest match. After flight data entered the same store, a city drive reported a mean of 337 km/h against a true 30. Air and ground are therefore stored and searched as separate partitions.

3.4 Direction

Turn angle comes from the gyroscope, absolute direction from the magnetometer:

$$\theta_t = \theta_{t-1} + \Delta\psi_{\text{gyro}} + \kappa(\psi_{\text{datum}} + \beta - \theta_{t-1}), \quad \kappa = 0.25 \quad (9)$$

where β is the *carry offset*: the angle between where the phone points and where the body or vehicle travels. Earlier versions learned β from GPS course during the first 180 s of a workout and then froze it. The current one uses no satellite information after the first point, in four parts.

A datum that cannot go singular. ψ_{datum} is not the platform’s heading. Core Motion reports the bearing of a direction fixed in the phone, and which direction depends on how the phone is held; when it points near vertical the reading is noise (§6). Instead, with the attitude $R = R_z(\text{yaw}) R_x(\text{pitch}) R_y(\text{roll})$ mapping the device frame to north–west–up, we take the bearing of whichever device axis lies most nearly horizontal — the one least aligned with gravity, switching only when another is flatter by 0.15 in $|\cos|$ — and carry the difference between the two axes across a switch so the reading stays continuous. Only offsets are ever learned against it, so its absolute zero does not matter.

The carry offset from cornering. In a turn a vehicle accelerates toward the inside of the turn. Writing the horizontal acceleration in the datum’s frame as a complex number $a = a_{\text{fwd}} + i a_{\text{right}}$, with v the estimated speed and ω the gyroscope turn rate (clockwise positive),

$$a_t \approx e^{i\beta} (i v_t \omega_t) \implies \hat{\beta} = \arg \sum_t \frac{a_t}{|a_t|} i \overline{v_t \omega_t} \quad (10)$$

over every riding second at $v \geq 4$ m/s. Normalising a_t keeps one pothole from outvoting a minute of corners. The acceleration must be the physical one, which is the *negative* of

what Core Motion reports as user acceleration (§6). The first estimate is made after 30 riding seconds.

Walking from the steps. With the phone in a pocket the compass was 56–137° from the direction of walking on five of the seven walks graded, so walking is steered by the steps instead. Over 4 s, the principal axis of horizontal physical acceleration is the line of travel, and its third moment along that axis is positive toward the front, because push-off is sharper than heel-strike. Each window’s vote enters a circular median of $(\text{vote} - \psi_{\text{datum}})$ over the last 20 windows, which picks the end of the current window’s axis; the heading is pulled toward it at 0.5 per second. A window counts only if the phone’s posture held within 15° across it, and settling at a new angle (25°) starts the record afresh. A phone lying flat ($|g_z| > 0.7$, in the hand) takes the end nearer the datum instead, because the vote came out reversed on the one flat walk recorded.

Redrawing. Every drawn point keeps the heading and offset it was drawn with. When a ride’s $\hat{\beta}$ moves by 3°, every point of the current ride is turned by the difference; once a walk has eight counted windows, its opening windows are redrawn with the settled median. The saved route therefore carries the best estimate reached, while the live display carries the one available at the time. A ride begins at 2.5 m/s, and at that moment the heading is set to $\psi_{\text{datum}} + \hat{\beta}$ outright, because sitting down turns the phone about a horizontal axis the gyroscope’s vertical rate does not see.

3.5 Projection

$$\Delta s = \hat{v} \Delta t, \quad \phi_t = \phi_{t-1} + \frac{\Delta s \cos \theta_t}{R}, \quad \lambda_t = \lambda_{t-1} + \frac{\Delta s \sin \theta_t}{R \cos \phi_t} \quad (11)$$

Only the first point of a route originates from GPS.

4 Results

Throughout this section *real* means the speed or distance GPS recorded at the same instant as the estimate, and *estimated* means what the app produced from the motion sensors alone. Real distance is the integral of GPS speed over time, not the sum of fix-to-fix jumps: on one 15 km journey those two differ by 35% (14,445 m against 19,496 m), because position scatter adds length that was never travelled.

4.1 Distance over a whole journey

Table 1 lists every journey. The best is a 15 km open road drive at +3.8%; the worst road case is a four-minute car-park trip at +61.6%. The last row is the phone carried in a pocket rather than resting in the vehicle, which is a different failure and is discussed in §8.

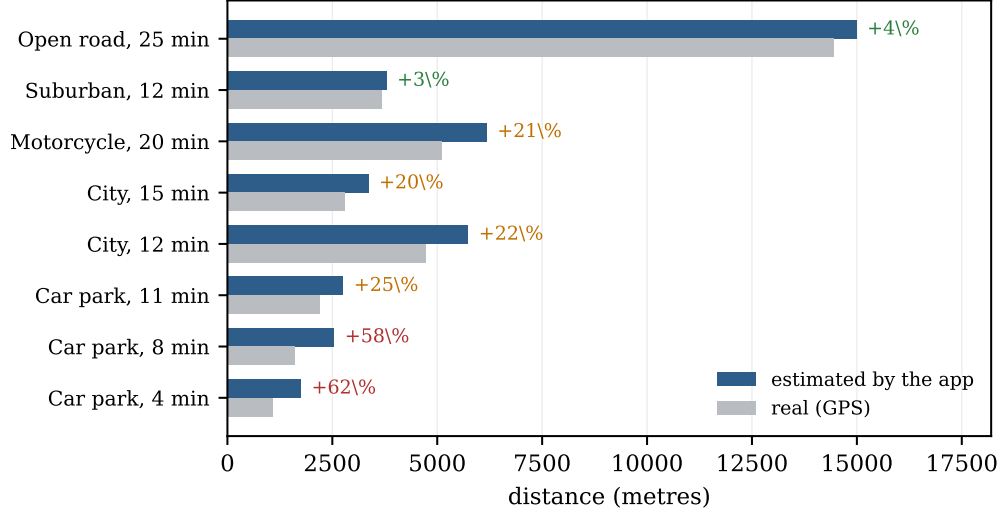


Figure 1: Estimated distance against real distance, eight journeys. The estimate is close when the vehicle is fast and steady, and reads long when it is slow.

Table 1: Every journey, in full. Average speed is over moving time only.

Journey	Estimated	Real	Error	Avg. speed
Open road, 25 min	14 994 m	14 445 m	+3.8%	43 km/h
Suburban, 12 min	3 793 m	3 678 m	+3.1%	28 km/h
Motorcycle, 20 min	6 181 m	5 118 m	+20.8%	35 km/h
City, 15 min	3 363 m	2 801 m	+20.1%	21 km/h
City, 12 min	5 741 m	4 720 m	+21.6%	29 km/h
Car park, 11 min	2 762 m	2 207 m	+25.2%	20 km/h
Car park, 8 min	2 548 m	1 616 m	+57.6%	20 km/h
Car park, 4 min	1 755 m	1 086 m	+61.6%	19 km/h
Motorcycle in a pocket	4 466 m	2 535 m	+76.2%	21 km/h

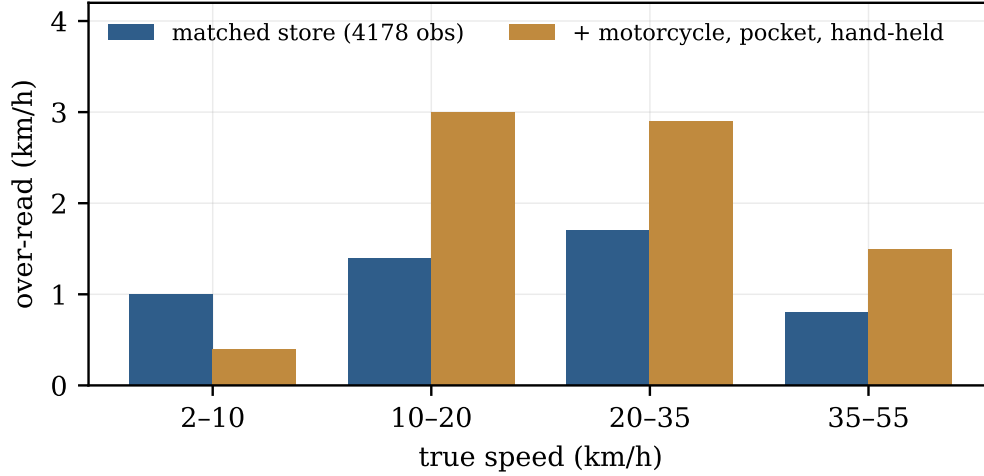


Figure 2: How much the estimate reads above the real speed, by real speed. Tested by hiding each stored example in turn and asking the model to recover it.

4.2 Why slow journeys read long

The ordering in Table 1 follows average speed, not any property of the route, and Figure 2 explains why. Tested against 4,178 stored examples from matched journeys — hiding each one in turn and asking the model to recover it — the estimate reads 1.0 to 1.7 km/h high at *every* speed.

A fixed error in km/h is not a fixed error in percent. Two km/h on a 50 km/h road is 2%; the same two km/h on a 5 km/h crawl is 40%. So the estimator that looks excellent on a motorway looks poor in traffic, without anything about it having changed.

This is also why no correction helps. Subtracting a constant 6 km/h — fitted on two journeys where the estimate happened to read exactly that much high — repaired those two and made three others worse, reducing total absolute error across all journeys by only 7%.

That account is correct as far as it goes, and it is not the whole story. It was measured by holding out stored examples, so it describes the error on the distribution the store holds. The riding has a different distribution, and §4.3 shows that the difference between the two — created entirely by the cache’s eviction rule — is the larger term.

4.3 The cache policy is the bias

The store holds 4000 observations. When it is full, an incoming observation displaces one drawn from the most crowded 2 km/h speed bin. The intent is to stop a rare 100 km/h example being squeezed out by thousands of red lights, and it achieves that. What it also does is make the store stop resembling the riding.

A query that lands between regimes is answered by whatever its twelve nearest neighbours

Table 2: What was ridden against what the store keeps, over 44 instrumented recordings. The eviction rule is the only difference between the two rows.

	< 5 km/h	5–30 km/h	> 30 km/h
Observations as learned	24%	31%	45%
Observations the store keeps	5%	26%	69%

happen to be, and after rebalancing most of them are fast. That predicts a specific signature of error: correct where the store is dense in both versions, wrong and high where the rebalancing has thinned the evidence. This is what the shipped application does — within a few percent above 50 km/h, roughly double at 15–30 km/h.

To test it rather than assert it, the estimator was reimplemented offline against the same feature extraction, the same K , the same decline threshold, and trained leave-one-ride-out on the other rides’ labelled windows. The only variable is how the training pool is assembled.

Table 3: Distance error, leave-one-ride-out. Rebuilding the pool under the shipped eviction rule reproduces the shipped error; the same estimator on an unbalanced pool does not. Weighting keeps the balanced store and corrects the imbalance at lookup.

Ride	Shipped app	Replay, evicted pool	Replay, natural pool	Replay, weighted
Motorcycle, 14 Sep	+32%	+7%	−5%	−4%
Motorcycle, 15 Sep	+53%	+23%	+3%	+8%
Motorcycle, 16 Sep a	+11%	−0%	−19%	−8%
Motorcycle, 16 Sep b	+39%	+38%	−2%	+10%
Motorcycle, 16 Sep c	+50%	+36%	+12%	+9%
Motorcycle, 16 Sep d	+30%	+15%	−9%	−8%
Mean error	36%	20%	8%	8%

Two things follow from Table 3. The estimator was never the problem: on a pool that was never rebalanced it is within 8% on the same rides, using the same eleven features that §5.4 had concluded carried no speed. And the fix does not require giving up the eviction rule — weighting each neighbour by $\pi(v)$ recovers the same 8% while keeping the rare fast examples an aircraft partition depends on. The residual gap between the replay’s 20% and the shipped 36% is the store’s older content, which the replay cannot reconstruct.

We record this as a modelling lesson of the same kind as the platform defects in §6: nothing about the eviction rule is wrong locally, it is sound cache management, and it was invisible for weeks because the store’s contents were never compared against the distribution they were drawn from.

4.4 Direction: the angle the phone sits at

The phone rarely points the way the vehicle travels. It sits at some angle in a cradle, a pocket or a cup holder, and that angle — β in §3.4 — has to be subtracted before the recorded heading means anything.

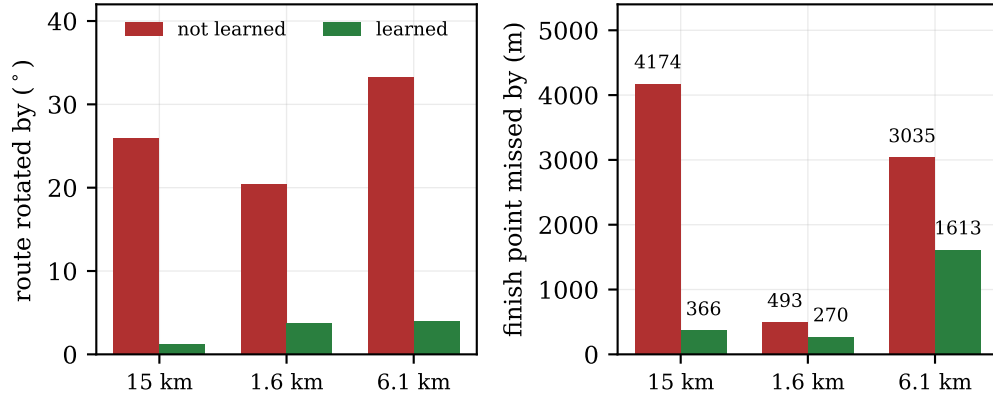


Figure 3: Why the angle matters, measured with the earlier GPS-learned version. The same journeys with the angle unlearned and learned. Left: how far the whole route is rotated. Right: how far the finish point lands from where it really was.

Figure 3 shows what that is worth. With the angle unlearned the route comes out the right shape and the right length and simply points the wrong way: on the 15 km journey the recorded straight-line displacement matched the real one to within 8 m while being rotated 26° , which put the finish 4,174 m from where it actually was. Learning the angle moves the finish to 366 m, 2.4% of the distance travelled. Those figures come from the earlier version, which measured the angle from GPS in the first three minutes and then held it.

Without GPS. The current version learns the angle from turns and uses no satellite information after the first point (§3.4). To see what that costs, the direction logic was replayed through every recording that logged the phone’s attitude: 61 recordings from 15 August to 24 September, car and motorcycle, mounted and pocketed, 476 minutes of riding over 344 km. Direction is replayed; distance is left as each build recorded it, so the routes below carry whatever speed error was current at the time. GPS enters only as the grader: for every half-minute of riding, the bearing between the GPS fixes 15 s either side — at least 150 m and four times their accuracy apart, each better than 60 m — against the circular mean of the drawn heading over the same half-minute.

Table 4 and Figure 4 summarise it. Without GPS the typical error is slightly worse than with the GPS-learned angle, 18.7° against 16.7° , but the failures are fewer: 72% of riding within 30° against 66%, and 3 recordings worse than 40° against 11. The GPS-learned angle is excellent when it is learned and wrong for the whole journey when it is not — frozen at a bad value from one poor fix, or never learned because the first three minutes

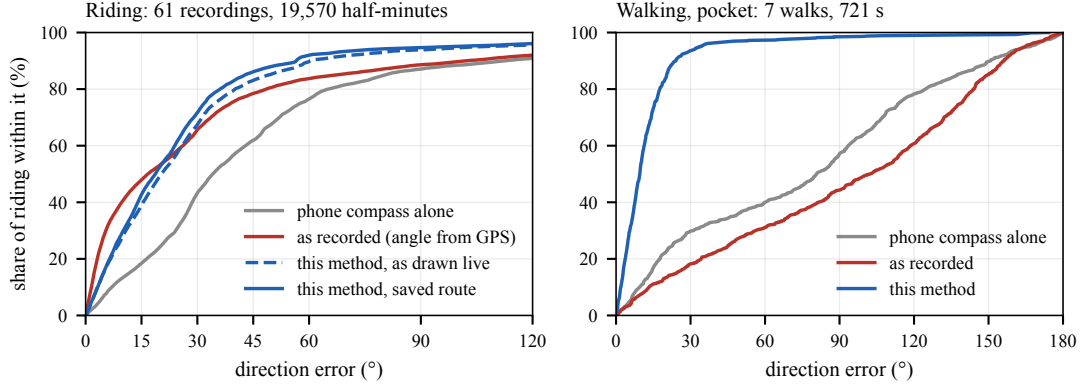


Figure 4: How far the drawn direction is from the real one. Left: 19,570 graded half-minutes of riding over 61 recordings. *As recorded* is what each build drew at the time, which for all but the last four recordings meant an angle learned from GPS. Right: 721 graded seconds of walking with the phone in a pocket, over seven walks.

Table 4: Direction error over every recording with attitude logged. Half-minutes of riding for the first four rows, seconds of walking for the last three. Pocket and flat split the 61 riding recordings by how the phone lay (35 and 26).

	Median	Within 30°			Recordings
	error	all	pocket	flat	over 40°
Riding, phone compass alone	34.1°	43%	30%	62%	24
Riding, as recorded (angle from GPS)	16.7°	66%	61%	72%	11
Riding, this method, live	20.3°	67%	68%	67%	7
Riding, this method, saved route	18.7°	72%	73%	69%	3
Walking, phone compass alone	80.8°	30%			
Walking, as recorded	102.3°	18%			
Walking, this method	9.6°	93%			

were spent standing still. The compass alone is the floor, and in a pocket it is a poor one: within 30° for 30% of riding. Walking changes the most, because the recorded builds had no working walking direction at all.

Against the angle measured from GPS on the 27 rides where that measurement is possible — measured with the platform’s heading as the datum, before the change to the axis heading — the cornering estimate was 11.0° off (median) after 30 riding seconds and 9.4° at the end of the ride, within 30° on 26 of 27. Redrawing each ride with its latest estimate is what moves the saved route ahead of the live one. It is not free: on about ten recordings the later estimate was the worse one, and redrawing spread it back over the ride, by up to 24° .

Figure 5 shows three recordings in full, and Appendix A shows all 61, each drawn against its own GPS track.

4.5 Underground car parks

Table 5: The two car-park journeys split by the ramp detector. A ramp is where the estimate fails; the road between the ramps is ordinary.

Journey	Where	Estimated	Real	Error
Car park, 4 min	on the ramp	267 m	67 m	+299%
Car park, 4 min	on the road	1 488 m	1 018 m	+46%
Car park, 8 min	on the ramp	618 m	229 m	+170%
Car park, 8 min	on the road	1 930 m	1 387 m	+39%

The two worst journeys in Table 1 are both car parks, and Table 5 locates the failure. A car crawling up a concrete spiral shakes as hard as one doing 40 km/h on a road, so the estimate on a ramp runs 170% to 299% long while the road between the ramps behaves ordinarily.

Detecting a ramp needs two signals together. The barometer alone flags every hill — 8 km of one 15 km journey, where the estimate is fine. Rotation alone flags every junction. A helical ramp is the only thing that climbs steadily *and* turns continuously the same way, so the app withholds distance when the barometer shows more than 0.08 m/s of sustained climb and the gyroscope more than 150° of same-direction turning within 30 s.

5 Negative results

We report these in full because each cost real effort and two share one cause.

5.1 Aircraft speed cannot be worked out at all

A flight was recorded with GPS valid on all 1323 ticks, reaching 706 km/h. Replaying it with GPS removed (Figure 6, Table 6) shows the problem clearly: signal is normally

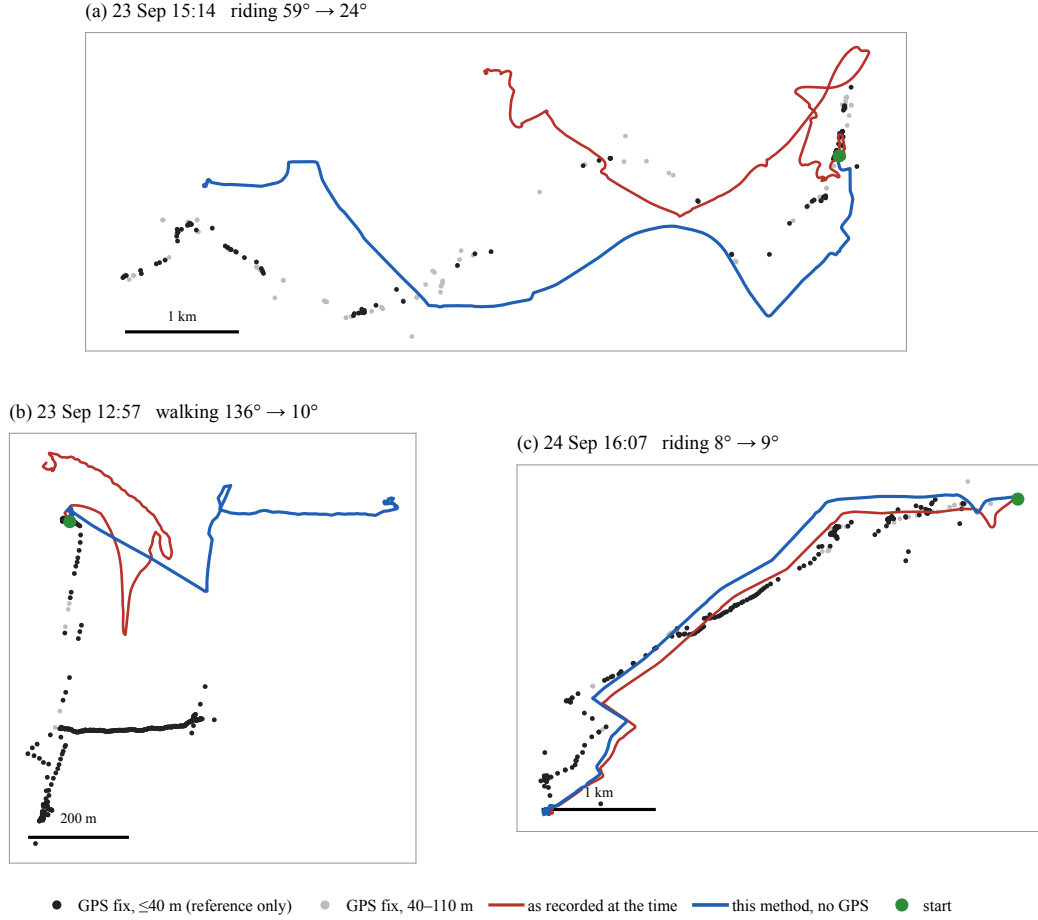


Figure 5: Three recent recordings, direction replayed without GPS, distances as recorded. Titles give the median direction error as recorded and with this method. (a) Phone head-down at 45° in a pocket, where the platform’s heading scattered 49° a minute (§6); recorded by a build that still used it. (b) A 600 m ride south and back, then a 270 m walk east: recorded by a build with no working walking direction, the walk was drawn 136° off. The ride in the replay is drawn worse than recorded: at 37 riding seconds its only estimate of the angle was poor, and the redraw carried it over the whole ride. It is also drawn short, a speed error of that build (§4.3). (c) The best recording to date.

Table 6: The same flight, in numbers. Signal is normally lost on the ground, which is the worst case: the last speed the app saw was taxi speed, and it holds it.

GPS lost	Speed held	Estimated	Error
At the gate	5.6 km/h	14.7 km	−82%
Just before takeoff	15.9 km/h	10.2 km	−87%
60 s into the climb	320.7 km/h	49.7 km	−31%
At cruise	307.6 km/h	46.5 km	−34%

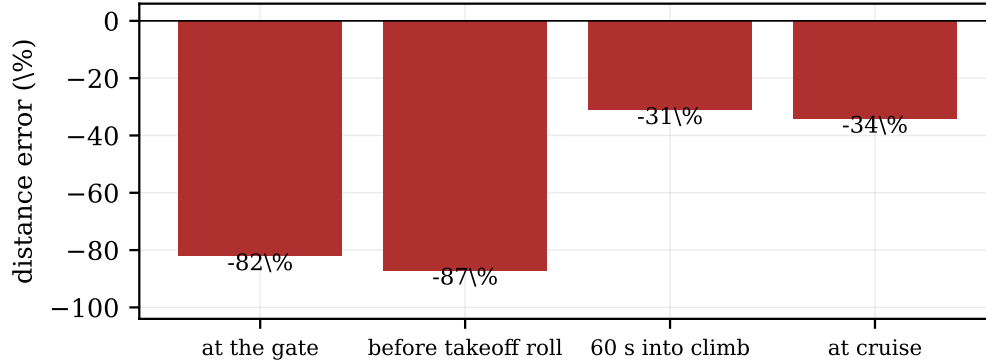


Figure 6: A real 22.8-minute flight replayed with GPS removed at four points, giving the distance the app would have recorded had signal been lost there.

lost on the ground, so the last speed the app has is taxi speed, and it holds it for the whole flight.

The correction that should rescue this — adding up the acceleration along the direction of travel — recovers about 60 km/h of a 679 km/h climb. After 400 seconds of held speed it had contributed -2 km/h. Direction, by contrast, is unaffected: 3.9° median across the same flight. The route points the right way and is the wrong length.

5.2 One cause, three failures

Core Motion’s attitude filter reinterprets sustained linear acceleration as a change in the gravity direction. The ambiguity itself is standard [15, 16]; what is reported here is its measured cost when the fusion is vendor-supplied and cannot be replaced. A takeoff roll is the most sustained acceleration a phone will ever experience, and is therefore precisely what the filter removes. This defeated two designs, and we had wrongly blamed it for a third:

1. *Aircraft speed integration*, above.
2. *Deriving β from acceleration* — withdrawn. A vehicle accelerates along its axis, so the world-frame acceleration bearing should give the travel direction; measured on three drives, circular concentration was $R = 0.15, 0.18, 0.19$, and we attributed that to the fusion. Most of it was ours. Core Motion’s user acceleration is the reaction the phone feels, the negative of the vehicle’s acceleration, so every bearing was reversed; and the frame was built by guessing between the attitude matrix and its transpose, which a phone lying flat cannot tell apart. Corrected, straight-line speed changes still give a poor answer — 44° median error — but the sideways acceleration of turns gives the carry angle to about 11° (§4.4).
3. *Inertial takeoff detection*. Integrating positive along-track acceleration and latching at 45 m/s in one unbroken episode should identify a takeoff and nothing else.

Replayed: today’s car 52.5 m/s (latched), a second car 209.9 (latched), a motorcycle 328.5 (latched), a two-minute *walk* 119.7 (latched), and the actual flight 15.4 — never latched. Backwards on every case. The episode rarely breaks in a vibrating vehicle because road noise holds the signal above the floor, so the integral accumulates the positive half of the noise indefinitely.

5.3 Regime identification does not transfer

Whole-session fingerprints — the mean normalised signature of a workout — separate regimes cleanly. Across eight sessions, same-vehicle-and-carry pairs sit 0.54–1.70 apart, same-vehicle-different-carry 1.81–2.87, and different-vehicle 4.30–6.97.

Restricting the neighbour search by that fingerprint nevertheless made prediction *worse* (+2.2 $\rightarrow$ +3.4 km/h). The ranking is unreliable at the individual level: for one car journey the nearest stored session was a car (1.38), but the second was a *hand-held* session (1.54) and the third a *motorcycle in a pocket* (1.74), both ahead of the same car driven the same morning (2.29).

5.4 On a motorcycle in a pocket the signature carries no speed at all

Every failure above is a matter of degree: the estimate is too high, or too low, or right about the wrong regime. This one is different in kind. Over a 19.1-minute motorcycle ride with the phone in a trouser pocket — 1146 ticks at a clean 1 Hz, Velocity Mode on for 1143 of them, three GPS fixes in the drawn route — the reported speed and the real speed are statistically unrelated: $R = +0.13$. The vibration signature and the real speed are related at $R = -0.02$, which is to say not at all.

Table 7 shows what that looks like from the outside. The estimate reports roughly 50 km/h whatever the motorcycle is doing, including when it is stationary at a light.

Table 7: Motorcycle, phone in a pocket. Mean estimated speed within each band of real speed. The estimate is flat: it does not fall when the motorcycle stops and does not rise when it accelerates.

Real speed	Ticks	Mean estimated
0–2 km/h (stopped)	47	36.2 km/h
2–10 km/h	25	54.4 km/h
10–25 km/h	3	57.8 km/h
25–45 km/h	14	55.5 km/h
45–70 km/h	54	49.8 km/h

The cause is visible in the signature itself. Its median value is 7.467 while the motorcycle is stopped and 7.386 while it is doing over 45 km/h — indistinguishable, and marginally higher at rest. A car carries the phone on a rigid surface and delivers road and tyre noise that scales with speed. A motorcycle delivers engine vibration, which tracks engine speed rather than road speed and is undiminished at a standstill in gear, through a layer of

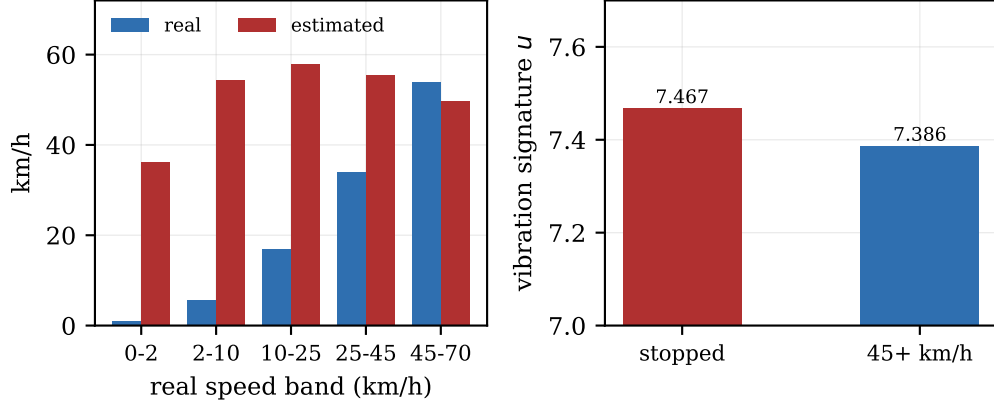


Figure 7: Left: estimated against real speed by band — the estimate is flat where the truth is not. Right: the vibration signature is indistinguishable between a standstill and 45 km/h, and marginally *higher* at rest.

clothing that damps what little road input survives. There is no speed information in the input, so no estimator built on this feature can recover one; the 47 stationary ticks accumulated 468 m of distance that did not happen.

What makes this the most serious result in the paper is not the error but the confidence. Fingerprint distance sat at 3.51 for the entire ride, outside the same-vehicle-different-carry range of 1.81–2.87 measured in §5.2 and approaching the different-vehicle range. The extrapolation flag — the mechanism whose job is to notice exactly this — was *false on all 1146 ticks*. The system reported 53.8 km/h at a red light and never signalled that it was answering about a regime it had never seen.

5.5 Correction: the pocket signature does carry speed

§5.4 is the strongest claim in this paper and it is partly wrong. The quantity measured there at $R = -0.02$ against real speed is a scalar from the superseded hand-built estimator — a scaled dominant vibration frequency, which the diagnostics file still records beside the learned model’s output — and not the eleven-dimensional signature §3.1 defines. The two were conflated. Re-measured on four motorcycle rides with the phone in a trouser pocket under open sky, the signal is present and strong:

Table 8: Motorcycle, phone in a trouser pocket, 16 September. RMS of vertical acceleration against real speed, in 4-second windows. The relationship the earlier scalar could not see.

Real speed (km/h)	0–5	5–15	15–30	30–45	45–60	60–90
Median RMS (m/s ²)	0.11	0.39	0.63	0.95	1.11	1.27

Over 331 windows on one ride, RMS ranks with real speed at Spearman +0.82, ahead of the shipped application’s own reported speed (+0.73); spectral centroid ranks at −0.64,

so the information is in amplitude rather than in where the energy sits. RMS is already one of the eleven features (f_{10} , §3.1). The flat ~ 50 km/h reading that §5.4 reports is therefore not an absence of signal in the input but the cache bias of §4.3 overwhelming it: the same features, searched against an unbalanced pool, place the same pocketed rides within 8–10%.

What survives from §5.4 is the part about confidence, which the correction makes worse rather than better. The system reported 53.8 km/h at a red light, the extrapolation flag was false on all 1146 ticks, and the diagnosis that followed — *this carry has no signal* — was itself wrong and stood for three weeks, because the log recorded a feature that was no longer the one being used. An instrument that reports a superseded quantity beside a current one, with no indication of which is which, produces confident wrong conclusions in the analyst as readily as in the application.

5.6 Confidence signals that do not work

Neighbour disagreement is the natural confidence measure for a k -NN regressor. It fails here in the wrong direction: out-of-distribution motorcycle queries had a *tighter* neighbour spread than in-distribution car queries (5.2 vs 12.4 km/h), because they land consistently in one wrong region. Session-level fingerprint distance correlated with speed bias at $r = -0.01$ over six journeys.

5.7 Discriminating a hijacked pedometer

On a motorcycle, engine vibration is counted as footsteps. Apple’s activity classifier reported `automotive` on 636 ticks and `walking` on 271 of one continuous ride, and each flicker allowed the pedometer to take over: it drove 27% of the ride, reporting a mean 3.3 km/h while GPS recorded 49.6.

Three per-tick discriminators were measured against real walking and rejected: vibration energy (79% caught at 21% false), gait-band spectral concentration (distributions overlap), and both combined with rotation rate (69% at 7%). What worked was history rather than instantaneous features: across two real walks the classifier *never* reported `automotive` in 190 stepping ticks, while on motorcycles the median gap since it last did was 84 s. Requiring two minutes of silence before a pedometer-driven revocation suppressed 61% of false hand-overs at zero measured cost.

5.8 Four more that were measured and rejected

Integrating between stops. Each detected stop is a known zero, so acceleration along the travel direction can be integrated between two of them with the drift pinned out at both ends. Measured over the segments of one ride: –65% of the real distance using the application’s heading, –50% using the raw motion heading, and –75% even when the GPS course was supplied as a perfect direction. Residual drift reaches 10–21 m/s per segment for a pocketed phone.

Time since the last stop. If the over-read were a pull-away artefact it would concentrate after a stop. It does not: on two rides the excess distance splits +64%, +54%, +48% and +81% across 10–30, 30–60, 60–120 and 120 s+ since the last stop.

Zeroing an idling engine. Shake does not separate a stationary motorcycle with its engine running from one crawling: accelerometer magnitude has median 0.22 m/s^2 stopped against 0.65 at 4–12 km/h, heavily overlapped. The best threshold tested removes 1142 m of phantom distance across 70 recordings while zeroing 312 crawling and 200 slow-riding ticks — about 2000 m that was really travelled.

Reweighting the amplitude features. Since f_{10} and f_{11} carry the speed information (§5.5), weighting them above the nine band energies in the distance metric should help. It changes nothing measurable (8% mean distance error at $\times 1$, $\times 2$ and $\times 3$). The neighbour search was never the problem; the pool was.

5.9 Six ways of finding direction that did not work

Each was built, replayed against GPS on the logged rides, and dropped. Numbers are median direction error unless stated.

The axis the motorcycle leans about. A leaning bike rolls about its own direction of travel, so the principal axis of horizontal rotation should give that direction up to a half-turn. It gives 25° even with the half-turn forgiven.

The thigh as a pointer. Walking, gravity in the phone’s frame runs down the thigh; seated, the thigh points forward, so the same device direction should point along the road. It does not: seated, that direction stayed $54\text{--}86^\circ$ from horizontal, and the estimate was 73° off.

A lean-proof forward vector. Leaning is a rotation about the forward direction, so solving for that direction in the phone’s own frame — least squares on $\mathbf{a}_{\text{dev}} = -v\omega [\mathbf{u}]_{\times} \mathbf{f}$ — should be immune to lean. It scored 19.9° against 15.9° for the planar method over 29 rides, better on six and worse on nine. Lean is not what limits the estimate.

Gating the compass by the gyroscope. Following the gyroscope and letting the compass move the baseline only while the two agree within 30° improved seven rides by more than 10° and worsened five; riding within 30° fell from 69% to 65%. It was treating a symptom whose cause was the heading definition (§6).

Switching headings by conditioning. Using the platform’s heading while its reference direction is well off vertical and the axis heading otherwise made the ride it was designed for worse, 72° to 78° : every hand-over copies the noisier source into the offset.

The thigh’s swing, for walking forward or back. A thigh swings forward faster than it returns, so rotation about the axis across the direction of travel should be skewed toward the front. It was wrong in 56–62% of two- to six-second windows, because the largest horizontal rotation of a pocketed phone while walking is about the direction of travel, not across it.

6 Systems failures

The first three defects below were not algorithmic, and each silently invalidated data for weeks.

A stale reading is not a missing reading. `CLLocationManager` suspends heading updates outside the foreground, and the cached property is never cleared. On an 81-minute drive the heading datum held *one value for 4790 of 4832 seconds* while the device physically rotated 5264°. A fallback to Core Motion’s heading had been added for the missing case and was never reached, because it was written as `clHeading ?? motionHeading` and a frozen value is not `nil`.

Background permission gating. `allowsBackgroundLocationUpdates` was enabled only under *Always* authorisation. Under *When In Use* the process was suspended mid-journey: a single 371-second hole in a 21-minute drive, during which the 50 Hz stream delivered 49 samples where 18,550 were due, accounting for the entire 20% distance shortfall.

Instrumentation that cannot express the failure. A 17-minute workout the user believed was recording without GPS had the mode active for 34 seconds; 63 GPS fixes entered the route. The guarantee held — the code path returns early — but nothing in the log stated the mode’s state, and it took an hour to establish from a side effect. Two columns now record it directly.

A repeated reading is not a new one. The diagnostics file carries the latest GPS speed, accuracy and position on every tick, holding the last fix until another arrives — in Velocity Mode, up to 45 s. One ride held a 0 km/h reading for 33 consecutive ticks while the rider pulled away, which scored as the estimator reporting 60 km/h at a standstill and put that ride at +47% where fresh fixes alone give +36%. The column was not wrong; it was undated. Every figure in §4.3 is computed against fixes under two seconds old, using an age column added afterwards.

Four meanings of “heading”. The direction work was held back for weeks by platform semantics rather than by the estimator. Core Motion’s heading is the bearing of a direction fixed in the phone, chosen by how the phone is held — fitted as $(-0.05, 0.87, 0.49)$ in the device frame on some rides and $(-0.02, 0.46, 0.89)$ on others — and when that direction points near vertical the reading is noise. On one ride the phone sat head-down at 45°, the direction was within 20° of vertical, and over each minute the heading scattered 49° around what the gyroscope measured; the route was drawn 72° off. The attitude itself was fine: the most nearly flat device axis scattered 1° over the same minutes. `CLHeading` follows yet another direction, 25–100° from Core Motion’s on a tilted phone, and the application handed over between the two whenever it entered or left the foreground; seven rides whose carry angle had looked like zero turned out to have been measured against `CLHeading`. Core Motion’s user acceleration is the reaction the phone feels, the negative of the vehicle’s acceleration: on pocketed rides it pointed 139–180° away from the acceleration GPS measured. And a 30 s running mean of the vertical

rotation rate, subtracted as gyroscope bias, is a high-pass filter that hands back every turn over the following minute — Core Motion’s rotation rate is already bias-corrected. The compass correction hides that last one, and removing it changed nothing measurable, so it remains.

A gate that cannot recover. The estimator refuses to answer when the workout’s fingerprint sits further than 3.0 from every learned session (§3.3). That fingerprint is the mean of the workout’s own GPS-labelled observations, and it only grows while GPS supplies a usable speed. On one ride GPS did so twenty times, during stops and a pull-away, and then effectively stopped: the fingerprint froze at 6.12 — a different-vehicle value, for the same motorcycle that had scored 0.29–1.88 that morning — and the gate refused every tick for 9.5 minutes, recording 20 m of a 4.3 km ride. Across 21 sessions the fingerprint exceeds 3 only while a session holds about twenty observations (3.35, 4.47, 4.78, each settling to 0.7–1.1) and never passes 2.57 after sixty. A refusal now requires sixty. The defect is not the threshold but that a quantity computed from an eight-sample minimum was allowed to be authoritative and could not be revised once the evidence feeding it stopped.

7 Limitations

- **Hand carry destroys the signal; pocket carry degrades it.** At matched speeds a hand-held phone reads 0.6–0.8 lower in the superseded summary feature, which stops varying with speed altogether. A pocket is different and was misdiagnosed for three weeks (§5.5): the signature there does track speed, and a pocketed ride places within 8–10% once the store’s composition is corrected for.
- **Stationary vehicles with running engines.** A motorcycle at a red light reads 5.2 km/h, accumulating 690 m over 339 stationary ticks. No tested signal separates an idling engine from a slow crawl.
- **Car-park ramps.** A vehicle crawling a concrete spiral shakes as hard as one at 40 km/h on a road; measured error is +169% to +297%. Detection requires two signals jointly — barometric climb above 0.08 m/s *and* signed net rotation above 150° over 30 s — since the barometer alone flags every hill and rotation alone every junction.
- **Straight ramps are not detected, and it matters less than expected.** The turn requirement excludes any ramp that does not spiral. The first straight one recorded — 12.6 m of climb over 1.15 km at 45–74 km/h, under 45° of net turning — showed why the rule should not simply be relaxed: the estimate there was already right (1135 m against 1153 m by GPS), so withholding it would have discarded a kilometre of correct distance. A turn-free climb test tightened to 6 m of rise at $R^2 \geq 0.9$ still fires on 2169 ticks across 64 recordings, almost all ordinary hills, and on 131 ticks of a flight’s cabin pressure. Only a *slow* straight ramp needs catching, and slow is the one thing the estimate cannot vouch for without GPS.

- **Unseen vehicles.** The model answers only for conditions it has observed.
- **A phone that moves in the pocket.** The carry angle is learned per ride and assumed to hold. Where the phone slowly rolled in a pocket during a ride the error grew to 30° , and redrawing a ride with a later, worse estimate spread it back over the whole ride. A walk that begins while the phone is still being put away is drawn from the compass until the phone settles.

8 Threats to validity

One device, one user, three vehicles. All 23 journeys were recorded on a single iPhone by one person, in one city, using one car, one motorcycle and walking. Every quantity here should be read as characterising that configuration. The transportation-mode literature moved to multi-user public datasets [6] precisely because single-subject results do not transfer, and we have no evidence ours do.

GPS as ground truth. Errors are measured against GPS recorded simultaneously. Where GPS is degraded — underground, and in the urban canyon immediately after — the reference is itself uncertain. For walking it can be worse than uncertain: on one 4.5-minute walk the Doppler speed averaged 1.06 m/s while position chords gave 1.6–2.2 m/s depending on the sampling interval, at 14–20 m accuracy. The accelerometer settles it without GPS — autocorrelation gives a 1.05 s stride period, so the application’s 1.55 m/s implies a 0.85 m stride and the Doppler figure implies 0.55 m, which is not the gait that was recorded. Walking figures in earlier drafts that rested on GPS alone should be treated as unproven. We use the integral of GPS speed rather than the sum of fix-to-fix displacements, because the latter accumulates position scatter: on one 15 km drive the two differ by 35% (14,445 m against 19,496 m), and the fix-summed figure is the wrong one. Segments with no valid GPS speed are excluded and reported separately, but a systematic bias in GPS speed at low speed would propagate into our low-speed figures undetected.

GPS as the direction reference. Direction is graded on half-minute chords between fixes that are often 30–60 m accurate, which puts a floor under every figure: every variant of the direction logic we replayed scored between 18° and 21° median error over all riding, however different its failures. Recordings worse than 40° , where the reference cannot be the explanation, are the more telling measure. Walking truth needs a straight half-minute of good fixes and exists for only seven walks.

No baseline comparison. We do not evaluate a published method on this data, so we cannot say whether a CNN trained on the same journeys would degrade the same way outside its distribution. The comparison we do make — matched store versus mixed store on identical held-out data — isolates store composition, not architecture.

Thresholds fitted on few journeys. The ramp detector (0.08 m/s climb with 150° signed rotation over 30 s) was selected by sweeping against seven journeys and choosing the lowest total absolute error. That is a small sample, and one journey regressed from -7.1% to -31.7% under the chosen setting because its ramp over-read had been cancelling an under-read elsewhere. We report it rather than tuning it away.

Data availability. The logs are per-tick CSV plus 50 Hz raw traces, and the results here are reproducible from them. They are not currently public. Releasing them would make this a labelled multi-vehicle, multi-carry vibration corpus, which to our knowledge does not otherwise exist; we consider that the most useful next step.

9 Conclusion

Vibration-derived speed is a sound basis for GPS-free route recording within its regime: fast, steady travel with the phone mechanically coupled to the vehicle, where we measure 3.8% distance error over 15 km. Direction no longer needs a satellite reference at all: learned from the vehicle’s own turns, and from the steps when walking, it is within 30° for 72% of riding and 93% of walking across every recording we have, against 43% and 30% for the phone’s compass alone. Outside that regime the failures are not gradual — a hand on the phone removes the signal entirely, and an aircraft’s defining acceleration is erased by the platform’s own sensor fusion.

We would emphasise three lessons over any result. The one added by this revision is that a component can be locally correct and globally biasing: the cache’s eviction rule is sound cache management, defensible in isolation, and it silently reshaped the distribution the estimator answers from (§4.3). It went unnoticed for weeks while four other explanations were measured and rejected, because the store’s contents were never compared with the distribution they were sampled from. Where a system learns continuously and stores what it learns, the storage policy is part of the model and belongs in its evaluation. First, the dominant error sources across this work were not modelling choices but sensor-platform semantics: a suspended process, a stale cached heading, an attitude filter making a defensible choice that happens to destroy the signal of interest. Second, an estimator that reports without expressing doubt is dangerous in a way that a wrong estimator is not; every failure above persisted for as long as it did because the system reported it with the same confidence as a correct reading. The shipped application now exposes its own error bars to the user.

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A Every recording, replayed

All 61 recordings with the phone’s attitude logged, in date order, each drawn with the current direction method and no GPS (blue) over its own GPS track (dots, used only for reference; black within 40 m, grey within 110 m). The start is green. Direction is replayed; distance is as each build recorded it, so a route that is the right shape but too long or too short is showing that build’s speed error, not a direction error. Titles give the date, how the phone lay, and the median direction error over the recording’s graded half-minutes.

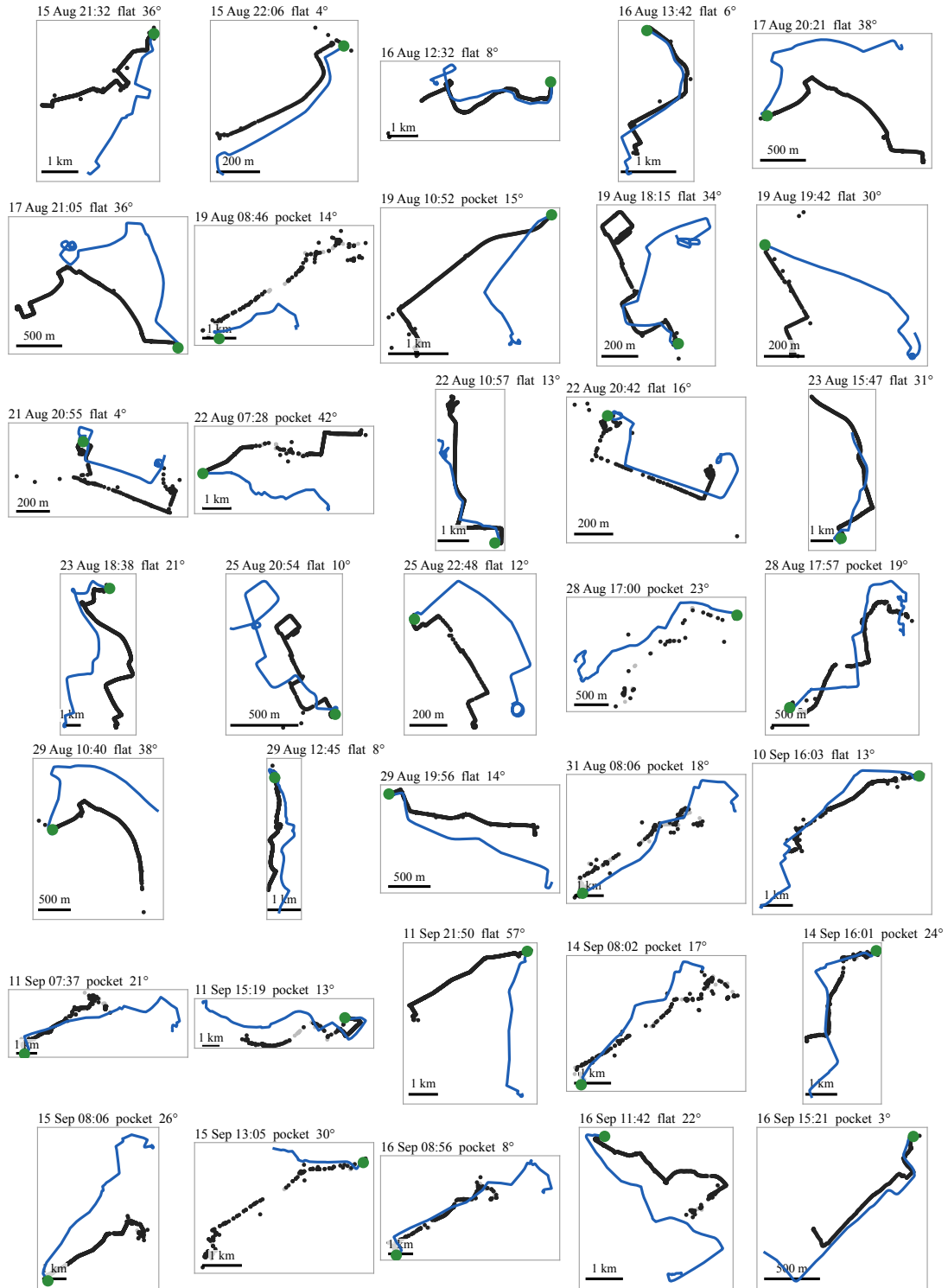


Figure 8: Recordings 1–35, 15 August to 21 September.

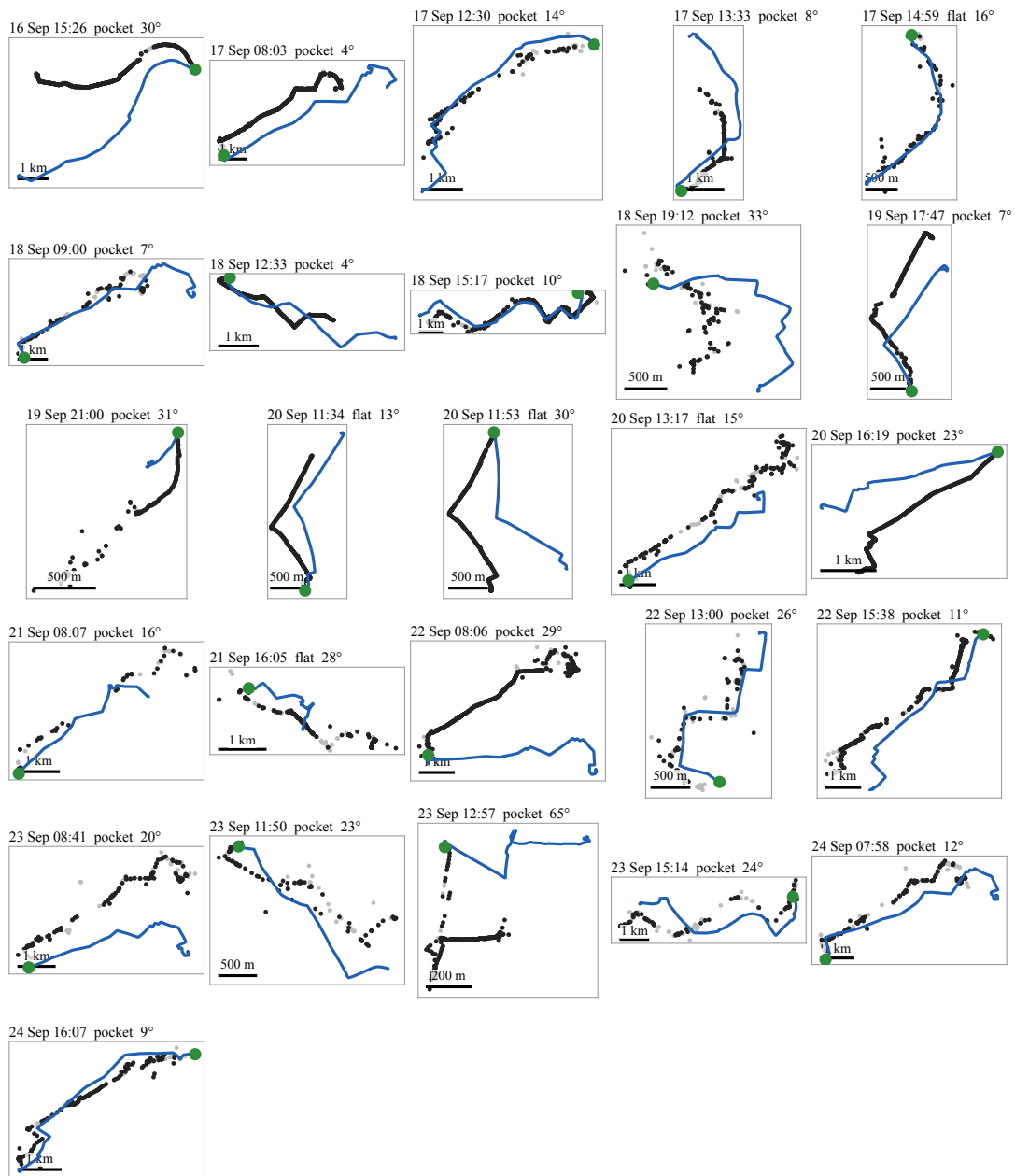


Figure 9: Recordings 36–61, 21 to 24 September.